Convex optimization - Project 5

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a)

$$\max \; C_{T+1}$$

$$\text{s.t.} \quad C_t \geq 0 \qquad t=1,\dots,T$$

$$C_{t+1} = C_t - \ell_t - F_t + G_t \qquad t=1,\dots,T$$

$$z_{tk} \geq 0 \qquad t=1,\dots,T, \quad k=1,\dots,K$$

$$z_{tk} = 0 \qquad t+M_k > T$$

$$F_t = \sum_{k=1}^{K} z_{tk} \qquad t=1,\dots,T$$

$$G_t = \sum_{\tau + M_k = t} z_{\tau k}(1+R_k)^{M_k} \qquad \tau \in \{1,\dots,T-1\}$$

b) Code is attached.